

University of Illinois at Chicago
Department of Chemical Engineering

CHE 382 Tuesday Section EIB 128
Chemical Engineering Laboratory II Syllabus

Instructor:

Alan Zdunek
Office: 150 EIB
Email: zdunek@uic.edu
Phone: 312-996-4607

Required Course Text:

Unit Operations of Chemical Engineering
W.L. McCabe, J.C. Smith, P. Harriott
7th Edition, McGraw-Hill, 2005.

Teaching Assistant:

Bailin Feng: bfeng24@uic.edu

Office Hours:

Instructor and teaching assistants will be available during the lab period (9am-12:30pm and 1:00pm-4:30pm) to help answer questions about the laboratory experiments, the experimental analysis, reports, and equipment maintenance.
Instructor and teaching assistants will also be available to answer questions by email or in-person by appointment.

Introduction:

CHE382, known as Unit Operations Laboratory II, provides students with a *hands-on opportunity to work with physical equipment in a laboratory environment*. Students, organized into lab groups, will write a pre-lab plan, conduct experiments, and write a final lab report. Bi-weekly, lab teams will make presentations to their colleagues and also present their last lab experiment in a final presentation at the end of the semester. There are also two safety assignments.

Lab Instruction

CHE 382 will be conducted **on-campus in EIB 128** following university guidelines for the safety and health of the students, instructor, and TA's. There will be no remote access during the lab periods; all students must be present in-person. Lab teams in each section will work on different experiments during each lab experiment cycle. Each lab experiment cycle takes two weeks and will consist of a prelab period and a final lab period. Lab teams will perform 6 lab experiments by the end of the semester following this two-week lab cycle.

List of Experiments

Each team will perform the following six experiments in CHE382:

Fluidized Bed	Process Control	Heat Transfer
Continuous Distillation	Liquid-Liquid Extraction	Gas Absorption

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Lab Cycle

Two lab periods are allocated for each experiment as follows, see Figure 1.

Period 1 (Lab Prep): This lab period is devoted to becoming familiar with the equipment and preparing a safety evaluation and experimental plan that will be carried out in Week 2. The team will review and understand the basic theory of the experiment, formulate a detailed step-by-step experimental procedure, and conduct a safety analysis on the equipment. A **Prelab Report** will be written by each team and is due the next day by uploading into Blackboard Assignments. See Blackboard for the Prelab Report template.

Period 2 (next week): This lab period is devoted to performing the experiment and collecting the data. The teams will analyze the results, formulate conclusions and recommendations, and investigate errors in measured variables. A **Final Lab Report** will be written by each team and is due the day before the next lab period. See Blackboard for the Final Lab Report template.

Next Lab Period 1: Each lab team will be assigned a new lab experiment to perform. At the beginning of the lab period, lab teams will give a presentation to the next lab team doing the experiment. The presentation template is in Blackboard, and the presentation will be submitted by the end of the lab period.

Week 1:	Jan	13	Introductions, Lab Team assignments	
Week 2:		20	Safety Assignment I Due	
Week 2:		20	Pre-Lab 1	LAB 1
Week 3:		27	Laboratory 1	
Week 4:	Feb	3	Pre-Lab 2	LAB 2
Week 5:		10	Laboratory 2	
Week 6:		17	Pre-Lab 3	LAB 3
Week 7:		24	Laboratory 3	
Week 8:	Mar	3	Pre-Lab 4	LAB 4
Week 9:		10	Laboratory 4	
Week 10:		17	Pre-lab 5	LAB 5
Week 10:		20	Safety Assignment II Due 3-20	
Week 11:		24	Spring Break No Lab Periods	
Week 12:		31	Laboratory 5	
Week 13:	Apr	7	Pre-Lab 6	LAB 6
Week 14:		14	Laboratory 6	
Week 15:	Apr	21	Class presentations (Final Lab Period)	
Week 16:		28	No Lab period	
Week 17:	May	4-8	FINAL EXAM WEEK (No lab)	

Figure 1. CHE 382 Laboratory Schedule for Tuesday AM and PM Sections

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Safety Assignments I and II: A safety assignment and quiz will be assigned through Blackboard assignment at the beginning of the semester to provide additional safety learning and comprehension. Students will also be required to complete one safety tutorial from the AIChE SACHE Certificate program on the AIChE website. AIChE membership is required but is free for undergraduate students. Specific details are in the Blackboard assignment.

Final Presentation: A final presentation of the last lab experiment will be given by each lab team to the whole lab section. The Presentation will be conducted in a classroom, room to be determined.

Blackboard: The Blackboard website is used for announcements, assignments and grades. It also contains the most up-to-date manuals for carrying out the various unit operations experiments (pdf files). Supplemental information on the experiments as well as report and presentation templates, technical writing tips, error analysis, and lab teamwork tips are provided in Blackboard to help aid in successfully preparing and carrying out the experiments, analyzing the data, and writing the reports.

Grade:

25% Lab Prep Report; 40% Final Lab Report; 5% Presentations,
10% Safety Assignment I & II; 20% Individual Performance Assessment

Late Assignments Policy:

Safety Assignment I & II: Up to 50% credit if turned in by next lab period, then 0%.

Lab Reports and Presentations: Up to 50% credit if turned in 1 day late. 0% credit after 1 day. One late submission up to 1 day late (prelab or final or presentation) without excuse will be allowed per semester.

Chemical Safety in the Laboratory

First and foremost, safety is an integral component in all laboratory environments, and certainly of all experiments and industrial processes. The Chemical Engineering Laboratory Safety document is your guidance in this regard. You must a) read, sign, and submit this document stating in effect that you have read, understood, and comply with the regulations contained therein. The safety document must be completed before you begin any lab work in this course. It

is a requirement that students in the lab take a proactive approach to safety and remind others of safety practices when necessary. The laboratory is an ideal place to develop safety awareness and skills; those who embrace the safety culture in the workplace will be highly regarded and rewarded professionally and financially in their future careers.

SAFETY GUIDELINES:

Eating and drinking are not allowed in classrooms.

Safety glasses must always be worn in the lab.

Safety glasses will be provided and must be worn at all times in the lab. Personal safety glasses will be given to each student and used throughout the year. They will be stored after each use in individual bins, identified with each student's name. Other PPE (personal protective equipment) such as disposable nitrile gloves, lab coats, and safety goggles will also be provided.

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Eating/Drinking and Safety Glass Compliance will be monitored by the instructor and TA during the lab period.

Non-compliance will be recorded. Two (2) instances of non-compliance will result in a zero on the Individual Assessment grade.

Leg Protection (No shorts) is required in laboratory.

Laboratory Access and Attendance

Student access to the Unit Operations Laboratory (128 EIB) will coincide with the scheduled class period; every Tuesday from 9:00am to 12:30 pm or 1:00pm to 4:30pm throughout the semester. The nature of laboratory studies is such that they are time and activity intensive; ***please be on-time so that the laboratory can begin promptly.*** The experiments are designed to be performed in two class periods (pre-lab and lab). If additional time is needed, permission from the instructor is required to complete or perform additional lab work.

Attendance

Lab attendance will be recorded for each lab period and will be one of the criteria used in the individual assessment grade. Arriving 15 minutes or later will result in a late mark (L). Arriving after 50 minutes or later will result in an absent mark (A) for that lab period. Please notify the TA or instructor before class if you cannot make the lab period. **One excused absence will be allowed and considered on an individual basis, but you must personally contact the instructor at least 24 hours before the lab period, otherwise it will be an unexcused absence. Email is best. Two (2) or more unexcused absences or four (4) Late marks will result in a zero on the Individual Performance grade.**

Teamwork

Students will be organized into lab teams. Each team will conduct experiments, write pre-lab and final reports, and present their work to their colleagues at the end of the semester. All team members are expected to participate jointly in planning and conducting the experiments, evaluating experimental data, analyzing results, writing reports and in the oral final presentation. The team should also decide on a team leader. Team member contributions will be tracked by each student filling out the Team Member Contribution form after each lab experiment (form is in Blackboard). The forms will be attached to the final lab report and will be used as part of the Individual Performance grade.

Academic Integrity -Student Community Standards

As a student and member of the UIC community, you are expected to adhere to the Community Standards of integrity, accountability, and respect in all your academic endeavors. When accusations of academic dishonesty occur, the Office of the Dean of Students investigates and adjudicates suspected violations of this student code.

Unacceptable behavior includes:

- Cheating
- Unauthorized collaboration
- Fabrication or falsification
- Plagiarism
- Multiple submissions without instructor permission

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- Using unauthorized study aids
- Coercion regarding grading or evaluation of coursework
- Facilitating academic misconduct.

Academic Integrity- continued

The above behaviors will not be tolerated and will result at a minimum in a zero for the report, presentation, or assignment and a zero on the Individual Assessment Grade. Other possible actions include an “F” on the final course grade, investigation, and disciplinary action by the Office of the Dean of Students and in egregious cases, expulsion from UIC.

Submitting Assignments and Plagiarism

By submitting your assignments for grading you acknowledge these terms; you declare that your work is solely your own, and you promise that, unless authorized by the instructor or proctor, you have not communicated with anyone other than your lab team on the prelab reports, final lab reports and presentations. Let's embrace what it means to be a UIC community member and together be committed to the values of integrity.

Plagiarism is defined as “the unacknowledged inclusion of someone else's words, structure, ideas, or data.” To avoid even the appearance of plagiarism, one must learn when and how to cite sources correctly. One helpful website is “Plagiarism.org”.

Here is further information on the subject from the UIC website:

Plagiarism: Definition and Policy

A student who submits a paper which in whole or part has been written by someone else or which contains passages quoted or paraphrased from another's work without acknowledgment (quotation marks, citation, etc.) has plagiarized. Maintain your integrity when completing assignments and be overzealous to give credit where it is due. If you ever are unsure about what constitutes plagiarism, ask your instructor.

Students who are found to have plagiarized work may be subject to various disciplinary actions, including a failing grade on a particular assignment, failure of the entire course, and possible expulsion from the university. In egregious cases, most instructors fail the student on the spot and file a complaint with Student Judicial Affairs.

For more information about violations of academic integrity and their consequences, consult the web site of the UIC Office of Student Judicial Affairs.